

UnReflectAnything: RGB-Only Highlight Removal by Rendering Synthetic Specular Supervision

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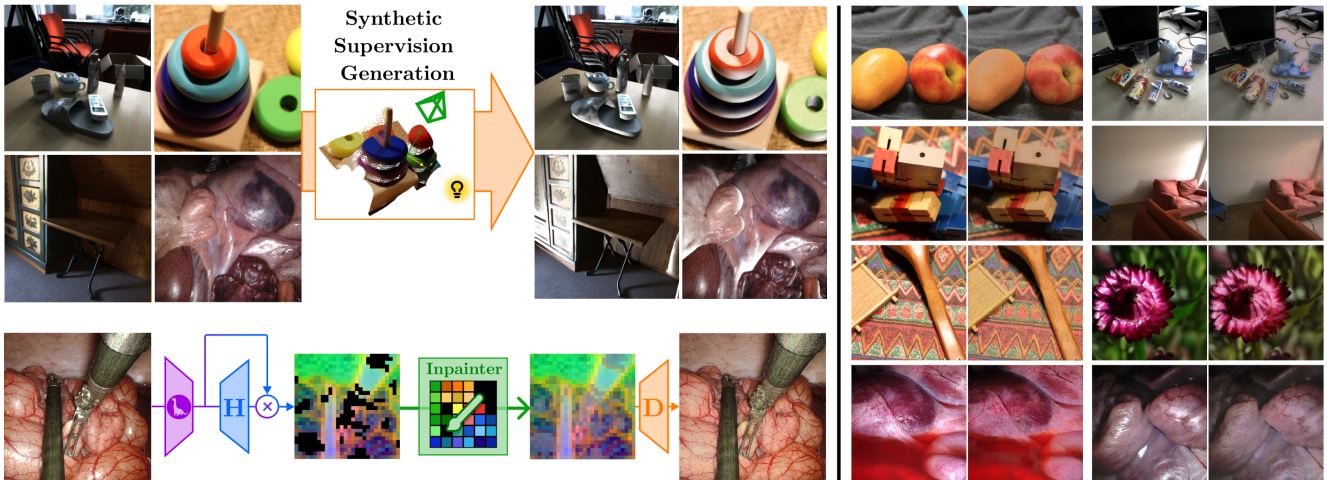


Figure 1. **UnReflectAnything** removes specular highlights from RGB images without paired supervision. A Virtual Highlight Synthesis pipeline renders physically plausible reflections from estimated geometry, providing realistic pseudo-pairs across natural and surgical domains. The model predicts a soft highlight map and inpaints masked DINOv3 tokens, reconstructing reflection-free diffuse images. It generalizes across diverse scenes, achieving faithful highlight suppression and texture recovery.

Abstract

Specular highlights distort appearance, obscure texture, and hinder geometric reasoning in both natural and surgical imagery. We present *UnReflectAnything*, an RGB-only framework that removes highlights from a single image by predicting a highlight map together with a reflection-free diffuse reconstruction. The model uses a frozen vision transformer encoder to extract multi-scale features, a lightweight head to localize specular regions, and a token-level inpainting module that restores corrupted feature patches before producing the final diffuse image. To overcome the lack of paired supervision, we introduce a Virtual Highlight Synthesis pipeline that renders physically plausible specularities using monocular geometry, Fresnel-aware shading, and randomized lighting which enables

training on arbitrary RGB images with correct geometric structure. *UnReflectAnything* generalizes across natural and surgical domains where non-Lambertian surfaces and non-uniform lighting create severe highlights and it achieves competitive performance with state-of-the-art results on several benchmarks. Project Page: alberto-rota.github.io/UnReflectAnything

1. Introduction

Specular highlights arise from mirror-like reflections on non-Lambertian surfaces, producing saturated, view-dependent artifacts obscuring scene content and corrupt downstream perception tasks such as segmentation, corre-

spondence, and photometric inference [11, 39]. In natural images, these highlights distort color fidelity and degrade restoration or relighting pipelines. In endoscopic scenes, bodily fluids, moist tissue and strong non-uniform lighting introduce intense specularities that occlude anatomy, bias texture cues, and hinder navigation and decision-making [7, 20, 21, 43]. Such artifacts critically impair depth estimation, optical flow, and stereo reconstruction in surgical robotics [7], where accurate scene understanding is essential and visual ambiguity can carry clinical impact.

Removing specular highlights from a *single RGB image* is fundamentally ill-posed: the specular component is saturated, spatially sparse, and entangled with geometry, material, and illumination. Polarization cameras can physically disentangle diffuse and specular reflections but their specialized hardware limits practicality [13].

To this end, previous work has extensively explored both classical and learning-based approaches for specular highlight removal using only RGB images. Classical methods rely on color priors [34], color ratios [32], or dichromatic reflectance models [31] to estimate diffuse components. However, these models often fail under complex materials or uncontrolled illumination. Deep learning methods [11, 39] learn diffuse-specular separation directly from data, but the scarcity of paired supervision and domain shift between synthetic and medical or other imagery limit generalization, often causing over-smoothing or hue distortions near highlight boundaries.

We introduce **UnReflectAnything** (Fig. 1), an RGB-only framework for single-image specular highlight removal across natural and surgical imagery. Our main contributions are:

- **Virtual highlight synthesis.** Realistic specularities rendering from monocular geometry using Fresnel-aware shading and randomized lighting, enabling supervision without paired diffuse-only data.
- **Token-space diffuse inpainting.** A transformer inpainter reconstructs masked DINOv3 patch tokens directly in feature space, restoring diffuse appearance with global context.
- **Hybrid geometry and pixel-level supervision.** A unified training scheme couples synthetic highlight rendering with token- and image-space losses, enforcing seamless boundary consistency and cross-domain robustness.

2. Related Work

Specular highlight removal is a long-standing challenge in computer vision. Early approaches relied on physical priors to suppress specular reflections. Tan *et al.* [34] and Shen *et al.* [32] leveraged color constancy and color ratio cues to decompose diffuse and specular components, while the dichromatic reflection model [31] offered a principled for-

mulation under simplified reflectance and illumination assumptions. Hardware-based techniques such as multi-flash imaging [10] or polarization cameras [22] enable more explicit separation of reflection components, yet their specialized setups are often impractical.

Learning-based methods have rapidly advanced specular highlight removal through large-scale data and task-coupled architectures. HighlightNet [11] established the first large-scale dataset of synthetic and real images with joint highlight detection and removal supervision. SpecularityNet [39] leveraged cross-polarization capture and adversarial training. To mitigate the need for paired data, MG-CycleGAN [17] introduced soft highlight masks and cycle-consistent adversarial training, enabling unpaired learning. More recently, DHAN-SHR [14] employed local-global hybrid transformers to achieve strong diffuse-specular decomposition without explicit mask supervision. Diffusion-based approaches have also emerged for specular suppression. StableDelight [1], inspired by StableNormal [41], leverages a generative diffusion framework trained on large-scale synthetic and real data to remove reflections and recover surface details.

While these approaches have shown strong performance on natural images, their adaptation to medical imagery presents unique challenges. Endoscopic scenes often contain broad, high-intensity reflections caused by moist tissue and dynamic lighting, making standard RGB-based solutions less reliable. To handle such domain-specific characteristics, several works have been proposed for endoscopic applications. Seminal efforts explored robust PCA-based decomposition [23], partial convolutions for specularity inpainting [43], and attention-driven architectures such as Endo-STTN [7], which employs temporal transformers to propagate texture across neighboring frames and enhance realism in surgical videos. However, Endo-STTN relies on binary highlight masks as explicit input queries, which are often unavailable in practice. Despite recent progress, paired surgical datasets remain scarce and existing models still struggle with severe saturation and complex illumination commonly encountered in endoscopic scenes.

Polarization data inherently disentangle specular and diffuse components, providing a powerful physical prior for both training and inference conditioning in learning-based highlight removal. SHM-GAN [3] employed polarized supervision to enable RGB-only inference, while PolarAnything [44] (PA) synthesized polarimetric cues (DoLP/AoLP) directly from RGB inputs, enabling polar-aware processing without specialized sensors. Polar-Free [40] (PF) leveraged diffusion priors with large-scale RGB-polarization pairs to enhance glare suppression, although it still depends on polarized inputs during inference. While effective, these approaches introduce additional sensing or calibration requirements that limit their practicality in

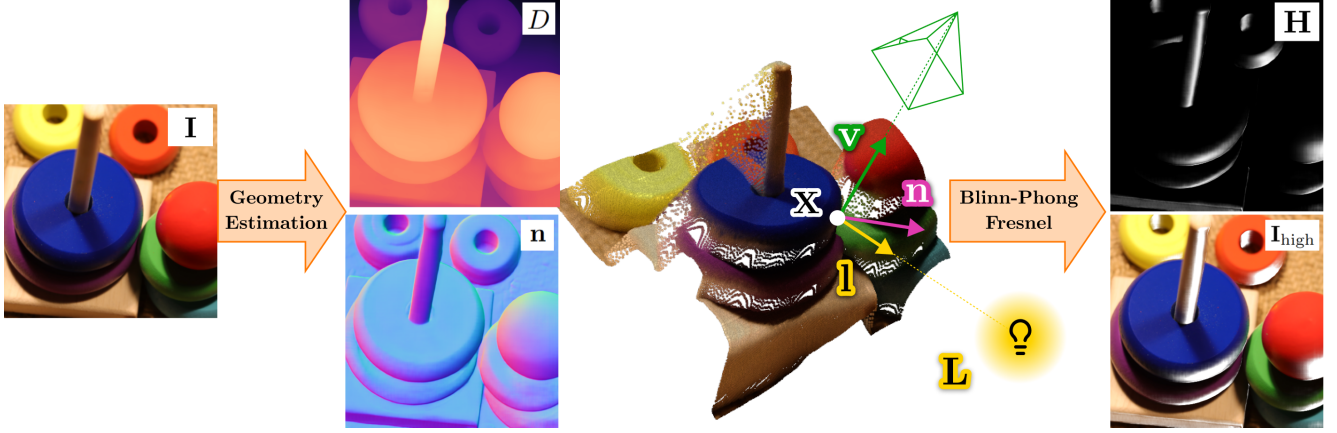


Figure 2. **Synthetic Highlight Generation Pipeline from any image.** Given a single RGB image (left), a per-pixel depth and surface normal are first estimated using a monocular geometry network. The recovered geometry defines a 3D point cloud $\mathbf{X}$ with associated surface normals $\mathbf{n}$ and view directions $\mathbf{v}$. The light source position $\mathbf{L}$ is sampled in camera coordinates, producing local illumination vectors $\mathbf{l}$. These geometric quantities drive a physically based Blinn-Phong [6] rendering mode generating a synthetic highlight intensity map that is photometrically consistent with the inferred scene structure. The highlight is finally composited with the input RGB.

compact or sensor-constrained imaging systems.

To our knowledge, *UnReflectAnything* is the first work to combine monocular geometry, Fresnel-aware specular rendering, and randomized lighting to generate physically plausible paired supervision for RGB-only highlight removal across both natural and surgical imagery.

3. Methodology

We integrate physically-grounded virtual highlight synthesis with token-space inpainting to recover reflection-free images from a single RGB view. The pipeline couples geometric rendering, highlight localization, and feature-level reconstruction into a unified framework.

3.1. Virtual Highlight Synthesis

Accurate 3D geometry estimates enable synthesis of realistic highlights on any RGB image for training supervision. Given $\mathbf{I} \in [0, 1]^{H \times W \times 3}$, we generate physically-plausible synthetic specularities by (i) inferring scene geometry from a single view, (ii) sampling a point light in camera space, and (iii) rendering a Blinn-Phong lobe with Fresnel modulation (cf. Fig. 2).

Monocular Geometry. We estimate metric depth $D \in \mathbb{R}_+^{H \times W}$, surface normals $\mathbf{n} \in \mathbb{R}^{H \times W \times 3}$, and intrinsics $\mathbf{K} \in \mathbb{R}^{3 \times 3}$ using an off-the-shelf method. For each pixel in homogeneous coordinates $p = (u, v, 1)^\top$, we back-project into its corresponding 3D camera coordinates $\mathbf{X}$ with $\mathbf{X} = D(p) \mathbf{K}^{-1} p$. The position of a point light $\mathbf{L} \in \mathbb{R}^3$ is sampled in 3D camera coordinates. We define the normalized view and light directions as

$$\mathbf{v} = \frac{\mathbf{X}}{\|\mathbf{X}\|}, \quad \mathbf{l} = \frac{\mathbf{L} - \mathbf{X}}{\|\mathbf{L} - \mathbf{X}\|}. \quad (1)$$

Specular Core. Let $\mathbf{h} = \frac{\mathbf{l} + \mathbf{v}}{\|\mathbf{l} + \mathbf{v}\|}$ be the half-vector between the light and view direction. For each point $\mathbf{x}$, we compute the Schlick-Fresnel [30] reflection coefficient

$$R = R_0 + (1 - R_0)(1 - \mathbf{v} \cdot \mathbf{h})^5, \quad (2)$$

where R_0 approximates the Fresnel reflectance at normal incidence. We employ the Blinn-Phong [6] reflection model to compute the per-pixel specular highlight intensity

$$\mathbf{H} = K_H R (\mathbf{n} \cdot \mathbf{h})^S, \quad (3)$$

where $K_H > 0$ controls the global highlight intensity and S quantifies the surface shininess. At training time K_H , S , and the light position $\mathbf{L}$ are sampled from empirically tuned uniform distributions that produce perceptually realistic highlight patterns, thereby increasing the heterogeneity of the supervision signal. We use a point light rather than a directional source to capture distance-dependent intensity falloff and localized specular effects — properties that are especially important for endoscopic illumination where the light source is at close range.

RGB Compositing. To obtain the final highlight-augmented image, the rendered specular lobe is *alpha-composited* onto the raw RGB input. We treat the highlight intensity H as an additive alpha mask scaled by a global scaling factor K_H , and blend it with the original pixel $\mathbf{I}$:

$$\mathbf{I}_{\text{high}} = (1 - \mathbf{H}) \mathbf{I} + \mathbf{H} (\mathbf{I} + K_H \mathbf{1}_3). \quad (4)$$

A critical aspect of training is that the original input $\mathbf{I}$ is *not* guaranteed to be highlight-free. On the contrary most images natively exhibit real specularities, especially

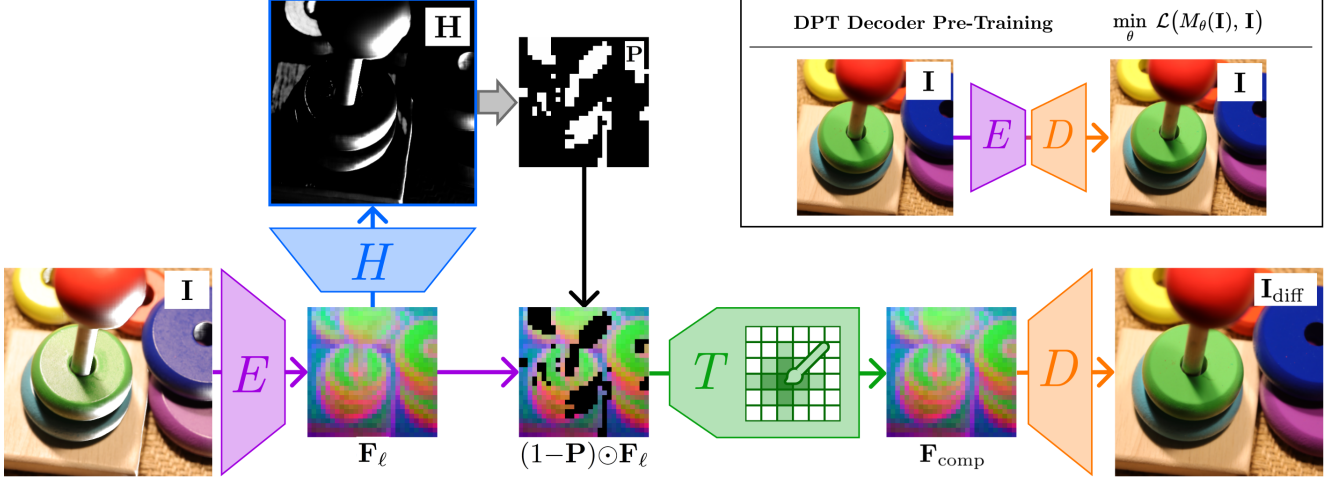


Figure 3. **UnReflectAnything model architecture.** A pretrained **DINOv3 encoder backbone** E extracts a hierarchy of multi-scale patch features from the input image (only the last feature map in the hierarchy is shown for clarity). A DPT-inspired **highlight predictor** H produces a soft, pixel-level highlight map, serving as a mask on the feature maps. A lightweight ViT-based **Token Inpainter** T operates on these masked features, learning to reconstruct the underlying diffuse features in place of those corrupted by highlights. An **RGB DPT decoder** D transforms the inpainted feature maps into a reflection-free diffuse RGB image. This decoder is pre-trained in an autoencoder-fashion to reconstruct the input RGB image from frozen DINOv3 features by minimizing the pixel-wise reconstruction loss $\mathcal{L}(M_\theta(\mathbf{I}), \mathbf{I})$.

endoscopy images. We denote these naturally occurring reflections as *dataset highlights*, hence specular regions already present in the raw data and not introduced synthetically. To differentiate them from our *synthetic highlights*, we detect *dataset highlights* via thresholding with a high luminance cutoff, τ_L , on the raw RGB images. Since supervising these pixels would mislead the network into interpreting saturated highlight regions as diffuse ‘white’ surfaces, we explicitly exclude all dataset-highlight pixels from supervision during training. The threshold is deliberately conservative: the resulting misclassifications consist predominantly of bright diffuse patches, which are then excluded from supervision. Rather than weakening training, this strengthens the supervision signal by steering the model toward higher-confidence patches. Compositing synthetic highlights onto images that may already contain real specularities is an approximation. We mitigate the resulting bias by randomly repositioning the light source across epochs, diversifying the supervision signal.

3.2. Model Architecture

Our model M inputs an RGB image $\mathbf{I} \in [0, 1]^{H \times W \times 3}$ and jointly predicts two outputs: a reflection-free diffuse-only image $\mathbf{I}_{\text{diff}} \in [0, 1]^{H \times W \times 3}$ and a score map for highlights $\mathbf{I}_{\text{high}} \in [0, 1]^{H \times W}$ that encodes the probability of specular regions: $(\mathbf{I}_{\text{diff}}, \mathbf{I}_{\text{high}}) = M(\mathbf{I})$.

M consists of four components (Fig. 3): an encoder E extracting hierarchical patch features, a highlight predictor H regressing a continuous probability map, a token inpainter T reconstructing masked feature regions cor-

responding to predicted highlights, and an RGB decoder D synthesizing the diffuse image from inpainted features. The overall process can be expressed formally as: $\mathbf{I}_{\text{high}} = H(E(\mathbf{I}))$ and $\mathbf{I}_{\text{diff}} = D(T(E(\mathbf{I}), \mathbf{I}_{\text{high}}))$. The highlight predictor guides the inpainter by localizing overexposed or specular areas, enabling the model to recover the underlying diffuse appearance before decoding it into the final RGB output.

Feature Extraction. We use a frozen DINOv3-Large [33] Vision Transformer as encoder E , which maps the input image $\mathbf{I}$ to patch-level feature tokens at four depths:

$$\mathcal{F} = \{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4\} = E(\mathbf{I}), \quad \mathbf{F}_\ell \in \mathbb{R}^{N \times C}. \quad (5)$$

These multi-scale tokens provide the hierarchical representations later consumed by the decoders for pixel-space reconstruction.

Highlight Prediction. A lightweight decoder H takes the multi-scale token set $\mathcal{F}$ and predicts a single-channel highlight intensity map $\mathbf{I}_{\text{high}} \in [0, 1]^{H \times W}$. This map identifies regions affected by specular reflections, providing a guidance signal for subsequent token inpainting.

Patch-Token Diffuse Inpainting. The token inpainter T (Fig. 4) reconstructs feature tokens corresponding to highlight-contaminated regions. Given the encoder features $\mathbf{F}_\ell \in \mathbb{R}^{N \times C}$ and a patch mask $\mathbf{P} \in [0, 1]^N$, T operates entirely in token space to infer the missing information and produce a coherent diffuse representation. Each of the following operations is performed separately at each ℓ layer,

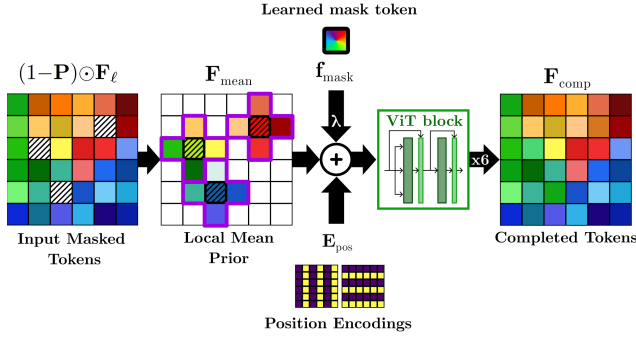


Figure 4. **Patch token inpainting logic.** For each token to be inpainted, a local neighborhood (purple borders) is used to compute the local mean priors. These mean priors are summed with Positional Embeddings and a learned mask token and fed into a sequence of transformer blocks to refine the tokens as final feature.

therefore we drop the ℓ notation for the rest of the section. $\mathbf{P}$ is obtained by spatially downsampling $\mathbf{I}_{\text{high}}$ to the encoder patch resolution using average pooling and thresholding the mean highlight intensity per patch.

Patch tokens to be inpainted are first substituted with a learnable *mask token* $\mathbf{f}_{\text{mask}}$, blended with a local mean prior $\mathbf{F}_{\text{mean}}$ computed from visible patch neighbors via depth-wise convolution and enriched by adding fixed 2D positional encodings $\mathbf{E}_{\text{pos}}$:

$$\mathbf{F}_{\text{seed}} = \mathbf{P} \odot [\lambda \mathbf{f}_{\text{mask}} + (1 - \lambda) \mathbf{F}_{\text{mean}}] + (1 - \mathbf{P}) \odot \mathbf{F} + \mathbf{E}_{\text{pos}}, \quad (6)$$

where λ is the local mean prior coefficient. The learnable mask token $\mathbf{f}_{\text{mask}}$ provides a semantic placeholder for missing content rather than defaulting to zeros [9]. Depth-wise convolution computes the local mean prior efficiently via channel-wise neighborhood averaging. Positional encodings are added (rather than concatenated) following standard ViT practice [8] to restore spatial awareness after token substitution. The *seed* sequence is then refined through a stack of six vision transformer layers performing self-attention over both visible and masked tokens. The final completed representation merges the inpainted and original visible tokens:

$$\mathbf{F}_{\text{comp}} = \mathbf{P} \odot \text{ViT}(\mathbf{F}_{\text{seed}}) + (1 - \mathbf{P}) \odot \mathbf{F}. \quad (7)$$

The resulting multi-scale set $\mathcal{F}_{\text{comp}} = \{\mathbf{F}_{\text{comp}}\}$ encodes highlight-free features that preserve both spatial structure and semantic coherence, and is subsequently decoded by D to reconstruct the diffuse image $\mathbf{I}_{\text{diff}}$.

Diffuse Image Reconstruction. We utilize a lightweight decoder to reconstruct diffuse RGB images, $\mathbf{I}_{\text{diff}}$, from completed multi-scale tokens, $\mathcal{F}_{\text{comp}}$.

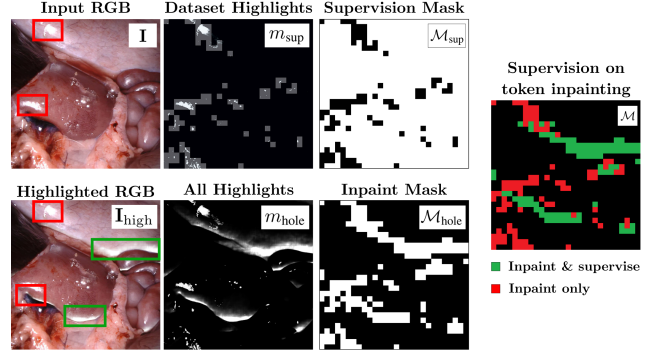


Figure 5. **Supervision masks for the token inpainting module.** Reliable supervision excludes *dataset highlight* regions (red). The inpainter must, however, learn to complete *all* highlight regions, including both synthetic (green) and dataset highlights.

3.3. Supervision

Our model is trained under a hybrid supervision scheme combining synthetic highlight annotations, pixel-space signals, and token-level objectives. The primary source of supervision is the *Synthetic Highlight Generation* pipeline, which enables pairing any RGB image $\mathbf{I}$ with a soft highlight mask $\mathbf{H}$ and its corresponding synthetically highlighted image $\mathbf{I}_{\text{high}}$.

Training relies on two pixel masks: (i) a *supervision mask* $m_{\text{sup}} \in \{0, 1\}^{H \times W}$ identifying pixels where ground truth is trustworthy (*i.e.*, regions free of inherent *dataset highlights*), and (ii) a *hole mask* $m_{\text{hole}} \in \{0, 1\}^{H \times W}$ denoting all regions to be inpainted, formed as the union of synthetic and dataset highlights.

For token-space supervision, both masks are downsampled to the patch grid (stride p), yielding the binary patch sets $\mathcal{M}_{\text{sup}} \subset \{1, \dots, N\}$ and $\mathcal{M}_{\text{hole}} \subset \{1, \dots, N\}$, which indicate patches eligible for reliable supervision and patches requiring inpainting, respectively.

Highlight Supervision. The highlight-prediction head H is trained using a weighted sum of soft Dice [4], L1, and Total Variation [29] losses as $\mathcal{L}_H = w_{\text{dice}} \mathcal{L}_{\text{dice}} + w_{\text{L1}} \mathcal{L}_{\text{L1}} + w_{\text{TV}} \mathcal{L}_{\text{TV}}$. Soft Dice [4] optimizes region overlap and is robust to class imbalance since highlights occupy only a small fraction of each image; L1 [24] enforces pointwise intensity accuracy; and TV [29] penalizes spatial noise, encouraging smooth predictions.

Token-Space Inpainting Supervision. The token inpainter predicts cleaned diffuse-only features for all highlight-affected patches, including both synthetically inserted and dataset-originating highlights. Because dataset highlights correspond to saturated or clipped intensities, their tokens do not constitute meaningful targets. Thus, we restrict supervision to the intersection $\mathcal{M} = \mathcal{M}_{\text{hole}} \cap \mathcal{M}_{\text{sup}}$, which identifies the subset of inpainted patches equipped

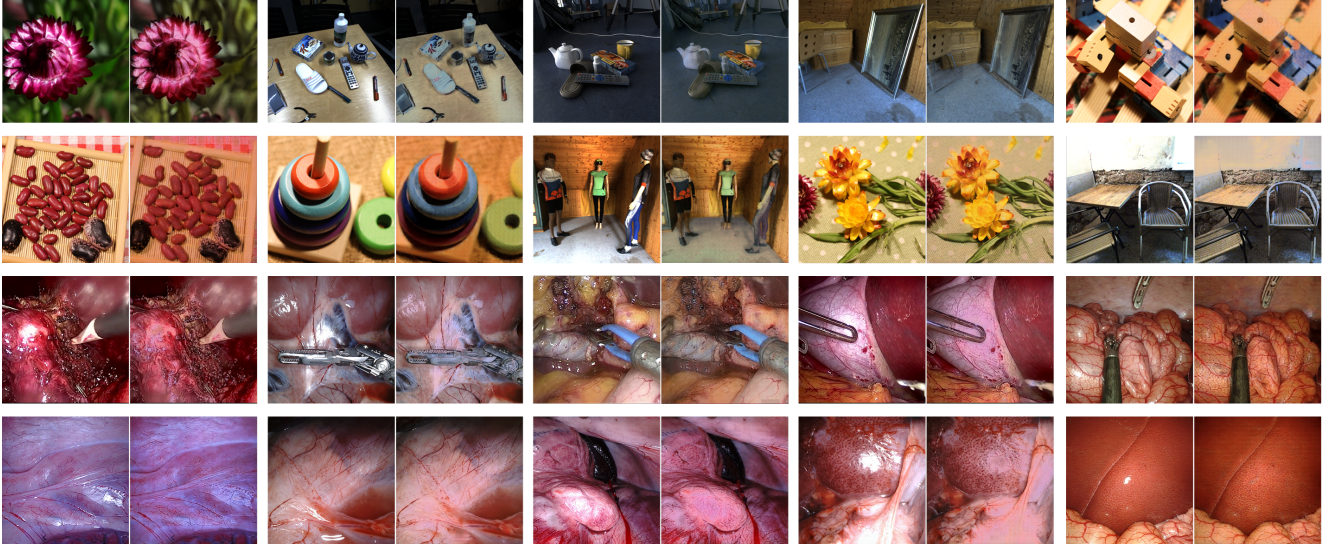


Figure 6. **Qualitative examples** for pairs of raw images with highlights (left) and their *UnReflectAnything*-processed counterpart (right). Our framework consistently removes, inpaints or attenuates specular highlights from images in several different domains.

with reliable ground truth. Fig. 5 provides a visual interpretation. For each patch $i \in \mathcal{M}$, the ground truth token is taken from the original clean image, $\mathbf{F}_i^* = E(\mathbf{I})$, before the synthetic highlight pipeline is applied. The inpainting loss utilizes cosine similarity and L1 loss, controlled by α :

$$\mathcal{L}_{\text{inp}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \left[\alpha \|\mathbf{F}_i^* - \mathbf{F}_i\|_1 + (1 - \alpha)(1 - \mathbf{F}_i^* \mathbf{F}_i^\top) \right]. \quad (8)$$

Decoder Pre-Training. Before inpainting is introduced, the RGB decoder D is *pre-trained* with a frozen DINOv3 encoder E in an auto-encoder fashion. The objective encourages the decoder to faithfully reconstruct the input RGB image from its DINOv3 features, with minimization of an L1 and SSIM term for RGB reconstruction:

$$\mathcal{L}_{\text{AE}} = \|D(E(\mathbf{I})) - \mathbf{I}\|_1 + (1 - \text{SSIM}(D(E(\mathbf{I})), \mathbf{I})). \quad (9)$$

This initialization ensures that D provides a stable and semantically meaningful feature-to-image mapping before the inpainting module is trained.

Decoder Fine-Tuning. After pre-training, the RGB decoder D is fine-tuned to restore seamless boundaries along inpainted regions, suppress residual highlights in the diffuse output $\mathbf{I}_{\text{diff}}$, and re-align feature statistics with the expected input distribution. To enforce smooth transitions at inpainting borders, *i.e.* seams, we define a thin boundary ring $r = \text{dilate}(m_{\text{hole}}) - m_{\text{hole}}$ and constrain discontinuities only within this region:

$$\mathcal{L}_{\text{seam}} = \|(\mathbf{I}_{\text{diff}} - \mathbf{I}) \odot r\|_1 + \lambda_g \|\nabla \mathbf{I}_{\text{diff}} - \nabla \mathbf{I}\|_1, r. \quad (10)$$

This term promotes color and gradient continuity across the inpainting boundary [45].

To prevent the diffuse decoder from reintroducing specular peaks, we additionally penalize overly bright pixels in $\mathbf{I}_{\text{diff}}$ using a smooth Charbonnier [42] loss. Let $B = \frac{1}{3}(I_R + I_G + I_B)$ and $m_{\text{hl}} = [B > \tau_m]$, we define:

$$\mathcal{L}_{\text{spec}} = \frac{1}{|m_{\text{hl}}|} \sum_{p \in m_{\text{hl}}} \sqrt{(B(p) - \tau)^2 + \varepsilon^2}. \quad (11)$$

We reintroduce the same RGB reconstruction loss used during decoder pre-training, this time comparing the predicted diffuse image $\mathbf{I}_{\text{diff}}$ with its reference $\mathbf{I}_{\text{diff}}^*$ while excluding dataset highlights via the supervision mask m_{sup} :

$$\mathcal{L}_{\text{RGB}} = \|\mathbf{I}_{\text{diff}} - \mathbf{I}_{\text{diff}}^*\|_1 + (1 - \text{SSIM}(\mathbf{I}_{\text{diff}}, \mathbf{I}_{\text{diff}}^*)). \quad (12)$$

4. Experiments

For the token in-painter, we use a sequence of ViT layers. For both highlight prediction head, H and diffuse RGB decoder D , we utilize a Dense Prediction Transformer (DPT)-based architecture [28].

Experiments run on 1× NVIDIA A100 (80 GB), batch size 32, 50 epochs, initial learning rate 5×10^{-4} decaying linearly every 10 epochs. We use MoGe-2 [38] for depth estimation, ViT layers for the token inpainter, and DPT [28] for both H and D .

We empirically set the approximation of the Fresnel reflectance at normal incidence, R_0 , to 0.04; the luminance threshold, τ_L , to 0.95; the local mean prior coefficient, λ , to 0.5; token-inpainting coefficient, α , to 0.25; and the mask

Table 1. Comparison across datasets that provide paired diffuse-only ground truth. The best score in each metric is highlighted in **bold**.

Method	PSD			SHIQ			SSHR		
	MSE _m ↓	PSNR ↑	SSIM ↑	MSE _m ↓	PSNR ↑	SSIM ↑	MSE _m ↓	PSNR ↑	SSIM ↑
DHAN-SHR [15]	0.006	24.863	0.868	0.001	33.815	0.982	0.001	37.318	0.971
PA+PF [40] [44]	0.050	14.101	0.645	0.089	12.009	0.604	0.024	18.102	0.432
SpecularityNet [39]	0.007	23.675	0.839	0.005	27.386	0.961	0.002	31.628	0.952
Fu <i>et al.</i> [12]	0.007	23.786	0.838	0.001	32.885	0.978	0.001	35.857	0.963
Li <i>et al.</i> [25]	0.019	18.819	0.716	0.007	24.351	0.919	0.004	26.898	0.909
EndoSTTN [7]	0.009	22.951	0.844	0.005	27.203	0.955	0.003	30.316	0.951
StableDelight [41]	0.007	23.384	0.838	0.007	22.844	0.921	0.001	32.838	0.886
OURS	0.004	17.230	0.911	0.003	17.610	0.988	0.002	28.415	0.971

Table 2. Comparison across datasets that do not provide paired diffuse-only ground truth. The best score in each metric is highlighted in **bold**. For all metrics, lower is better.

NATURAL DOMAIN						
Method	CroMo		HouseCat6D		SCRREAM	
	LSR	NIQE	LSR	NIQE	LSR	NIQE
DHAN-SHR [15]	0.063	0.084	0.224	0.142	0.073	0.075
PA+PF [40] [44]	0.160	0.103	0.280	0.167	0.206	0.093
SpecularityNet [39]	0.179	0.070	0.192	0.141	0.109	0.070
Fu <i>et al.</i> [12]	0.129	0.068	0.241	0.132	0.147	0.059
Li <i>et al.</i> [25]	0.113	0.088	0.184	0.143	0.088	0.085
EndoSTTN [7]	0.028	0.095	0.078	0.149	0.039	0.078
StableDelight [41]	0.176	0.073	0.318	0.142	0.270	0.078
OURS	0.012	0.061	0.033	0.120	0.002	0.056

SURGICAL DOMAIN						
Method	Cholec80		SCARED		StereoMIS-Tr	
	LSR	NIQE	LSR	NIQE	LSR	NIQE
DHAN-SHR [15]	0.115	0.101	0.114	0.131	0.077	0.165
PA+PF [40] [44]	0.111	0.161	0.110	0.162	0.151	0.191
SpecularityNet [39]	0.150	0.099	0.103	0.116	0.112	0.115
Fu <i>et al.</i> [12]	0.157	0.072	0.147	0.098	0.111	0.115
Li <i>et al.</i> [25]	0.035	0.108	0.021	0.151	0.038	0.134
EndoSTTN [7]	0.021	0.135	0.020	0.175	0.052	0.163
StableDelight [41]	0.316	0.046	0.262	0.103	0.278	0.110
OURS	0.002	0.100	0.011	0.144	0.022	0.131

threshold decoder fine-tuning, τ_m , to 0.85 with its corresponding stability bias, ϵ , to 10^{-6} . Cf. Sup. Mat. for details on loss weights and scheduling.

4.1. Datasets

In order to capture a wide range of domains, we utilize a combination of indoor (SCRREAM [18], HouseCat6D [19]), outdoor (CroMo [37]), and endoscopic (SCARED [2], Cholec80 [35]) datasets.

We group datasets by availability of diffuse ground truth. *Diffuse-referenced* (PSD [39], SHIQ [11], SSHR [12]) provide paired diffuse images $\mathbf{I}_{\text{diff}}^*$. *Non-referenced* (StereoMIS-Tracking [16], SCRREAM, HouseCat6D, CroMo, SCARED, Cholec80) contain only raw or polarized RGB. To our knowledge no public surgical dataset provides paired reflection-free ground truth.

Table 3. Comparison of each method’s impact on pixel-matching, evaluated using the epipolar error (E_{ep} , lower is better [↓]), and inlier ratio (I_R , higher is better [↑]). Best results are in **bold**.

NATURAL DOMAIN						
Method	CroMo		HouseCat6D		SCRREAM	
	E_{ep} ↓	I_R ↑	E_{ep} ↓	I_R ↑	E_{ep} ↓	I_R ↑
DHAN-SHR [15]	0.481	0.989	0.592	0.995	0.831	0.997
PA+PF [40] [44]	0.364	0.992	0.722	0.998	0.369	0.996
SpecularityNet [39]	0.370	0.993	0.710	0.987	0.620	0.994
Fu <i>et al.</i> [12]	0.532	0.997	0.976	0.998	0.386	0.997
Li <i>et al.</i> [25]	0.467	0.992	0.606	0.997	0.578	0.999
EndoSTTN [7]	0.326	0.996	0.952	0.996	0.867	0.999
StableDelight [41]	0.370	0.993	1.042	0.994	0.599	0.991
OURS	0.301	0.997	0.759	0.991	0.423	0.998

SURGICAL DOMAIN						
Method	Cholec80		SCARED		StereoMIS-Tr	
	E_{ep}	I_R	E_{ep}	I_R	E_{ep}	I_R
DHAN-SHR [15]	0.530	0.998	1.013	0.996	0.729	1.000
PA+PF [40] [44]	0.567	0.998	1.005	0.996	0.665	1.000
SpecularityNet [39]	0.587	0.995	1.155	0.996	0.213	1.000
Fu <i>et al.</i> [12]	0.437	0.997	1.155	0.997	0.958	1.000
Li <i>et al.</i> [25]	0.612	0.997	1.014	0.993	0.958	1.000
EndoSTTN [7]	0.530	0.997	0.804	0.988	1.922	0.974
StableDelight [41]	0.409	0.996	0.898	0.998	0.232	0.991
OURS	0.355	0.998	0.877	0.998	0.167	1.000

4.2. Evaluation on Diffuse-Referenced Datasets

For datasets that provide paired diffuse ground truth, we directly evaluate the reconstructed diffuse image $\mathbf{I}_{\text{diff}}$ against its reference $\mathbf{I}_{\text{diff}}^*$ using standard full-reference metrics (cf. Table 1), including mean squared error within the high-light masks (MSE_m), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index Measure (SSIM). Across the PSD, SHIQ, and SSHR benchmarks, our method remains consistently competitive and provides clear advantages in terms of structural fidelity. On PSD, *UnReflectAnything* delivers the strongest structural consistency, indicating that the model is particularly effective at restoring fine textures and attenuating residual highlight artifacts. On SHIQ and SSHR, while classical baselines perform with lower MSE_m and dominate in attenuating highlights, our approach yields

the most stable structural outcomes (SSIM), which is notable given the severe saturation patterns and limited texture cues characteristic of this dataset. Our model attains lower MSE_m but also lower PSNR, indicating it may be less effective at preserving fine details in the non-highlighted regions despite achieving lower global error.

4.3. Evaluation on Non-Referenced Datasets

For datasets that lack diffuse ground truth, we evaluate (cf. Table 2) perceptual fidelity using NIQE [27], and a Luminance Suppression Ratio (LSR). NIQE assesses adherence to natural scene statistics, while we define LSR as a measure of how effectively highlight intensity is reduced without global dimming (for definition cf. Supp. Mat.). Across the natural-domain datasets (CroMo, HouseCat6D, and SCR-REAM), our method consistently achieves the strongest highlight suppression according to LSR, while maintaining NIQE scores that are competitive with or superior to those of existing baselines. On surgical datasets, where specularities are more frequent and structured, our approach systematically attains the lowest LSR across Cholec80, SCARED, and StereoMIS-Tracking, indicating effective suppression of residual luminance in highlight regions, while remaining comparable to competing methods in terms of NIQE. These results suggest that the model can reduce specular artifacts without substantially degrading global perceptual quality.

4.4. Downstream Performance

We evaluate the influence of specular highlight suppression in a relative pose estimation task. For each frame pair, we detect DISK [36] keypoints, match them with LightGlue [26], and estimate the essential matrix $\hat{E}$ with MAGSAC++ [5], measuring symmetric epipolar error (E_{ep}) and inlier ratio (I_R) as metrics (cf. Table 3).

Across the natural-domain datasets, our method yields competitive results while achieving state-of-the-art results on CroMo. These findings indicate that the effect of highlight suppression varies across scenes, enhancing geometric alignment or correspondence stability depending on scene characteristics. In surgical datasets, where highlights are more intense and widespread, our model shows a more uniform advantage. It attains leading or joint-leading inlier ratios across Cholec80, SCARED, and StereoMIS-Tracking, and delivers competitive epipolar errors, including the lowest residuals on StereoMIS-Tracking. These trends indicate reducing specular contamination helps preserve keypoint localization under challenging clinical illumination, where reflections often interfere directly with geometric cues.

The downstream results demonstrate suppressing specular peaks improves both the robustness and consistency of geometric estimation, supporting more reliable correspondence matching in both natural and surgical environments.

4.5. Discussion

We present diverse predictions of *UnReflectAnything* in Fig. 6 and a qualitative comparison with the strongest SOTA baselines in the supplementary materials. Across natural and surgical domains, our framework consistently suppresses, inpaints, and attenuates specular highlights, including in challenging endoscopic scenes with frequent and spatially complex reflections. In contrast, diffusion-based methods often leave residual highlights, under-attenuate bright regions, or introduce local structural inconsistencies; in surgical images, they may also over-dim the entire frame. By design, our feedforward architecture favors conservative reconstruction over hallucination, which is desirable in medical settings where fidelity is critical and also makes real-time deployment more plausible than diffusion-based alternatives.

Despite strong cross-domain generalization, several limitations remain. The model is less reliable on transparent or refractive objects, where diffuse and specular components are difficult to separate. It can also reduce structure or resolution in non-highlight regions, suggesting that explicit edge constraints or high-frequency priors may be beneficial. More semantic reasoning could further help, since training-time highlight supervision relies on an ad hoc hard threshold and thus mainly exploits luminance cues. Finally, the token inpainting module T depends strongly on a binary highlight mask: with soft masking, where downsampled I_{high} directly provides patch weights, the model only weakly modifies soft highlight regions and often fails to remove them effectively (cf. supplementary materials), especially for gradual low-gradient highlights. Relaxing the dependence on luminance cues, both at training and inference time, would improve robustness to early-stage failures.

5. Conclusion

We present *UnReflectAnything*, an RGB-only method for single-image highlight removal that learns from synthetically rendered supervision grounded in monocular geometry. By coupling a physically motivated highlight synthesis pipeline with token-space inpainting, our method achieves faithful diffuse reconstruction and generalizes across diverse real-world domains including challenging endoscopic imagery, without requiring paired data. *UnReflectAnything* generalizes across diverse domains and achieves SOTA results on several benchmarks while maintaining strong overall performance, offering a solid foundation for RGB-only highlight removal with potential for clinical applications. Beyond visual improvement, *UnReflectAnything* enhances geometric consistency in downstream correspondence and camera-pose estimation tasks, setting a new direction for physically guided learning in reflection suppression.

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